Table 2.1 components of weight gain during pregnancy.

Gestation (weeks)	12	13-27	28-40
Foetus	5	1500	3000
Placenta and amniotic fluid	50	1000	1500
Maternal tissues and blood	600	6000	7000
Total weight gain	655	8,500	11,500

Source – Hytten, F. and Leitch, I. Physiology of Human Pregnancy, Blackwell Scientific Publishers, Oxford, London and Edinburgh, 1971.

Table 2.2 recommended weight gain for pregnant women based on body mass index.

Weight category based on BMI	Total weight gain (kg)
Under weight (BMI < 19.8)	12.5 – 18
Normal weight (19.8 – 26)	11.5 – 16
Over Weight (26 – 29)	7 – 11.5
Obese (>29)	6.0

Source – B.Srilakshmi.,Dietetics., New Age International Publishers.



Your heart grows! Yes, your heart organ actually enlarges while you are pregnant. It works harder and beats at a more rapid pace due to the increased volume of blood in your body. Your blood volume alone increases by 40 – 50% during pregnancy.

Due to undernourishment of the mother the baby is at an increased risk of being premature with low birth weight and development irregularities. Intrauterine nutrition is highly important for the growth of the central nervous system and kidneys of the foetus, which mature during the latter part of the pregnancy. Therefore nutrition deficits before birth can never be wholly reversed after birth.

Schematic diagram, figure 2.2 shows the relationship between maternal and foetal malnutrition.

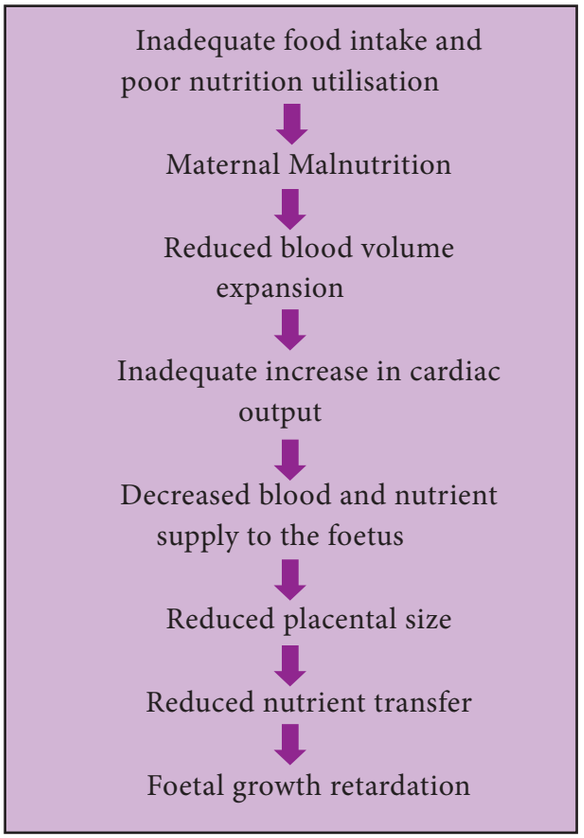


Fig 2.2 Relationship between maternal and foetal malnutrition.

2.1.2 Effects of under nutrition on the mother

a) Maternal body size

Early marriage tradition in many poor societies and pregnancy during adolescence before the genetic potential of growth is achieved, imposes additional burden which results in the poor growth of the foetus and birth of growth retarded child.

b) Multiple nutrient deficiencies.

Multiple nutrient deficiencies is due to poor diet are markedly aggravated during

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